

Trustworthy Cities: Ethical Urban Artificial Intelligence

By Marta Galceran, Research Fellow - Global Cities Programme, Barcelona Centre for International Affairs (CIDOB)

This module explains how cities are increasingly embracing the use of algorithmic tools to address all sorts of urban challenges and improve the provision of public services. It also discusses the ethical implications of the use of artificial intelligence in urban contexts and, building on the research of the Global Observatory of Urban Artificial Intelligence, it provides an overview of some of the initiatives and ethical principles promoted by cities.

Lecture Transcript

0:05 Hello everyone, and welcome to this new module of the course AI Ethics: Global Perspectives. My name is Marta Galceran, and I'm a research fellow at the Global Cities Program at CIDOB. For those of you that might not be familiar with this institution, we are the Barcelona Centre for International Affairs. Next year, we are turning 50, which means that CIDOB is a well established and important institution in Catalonia. Yet the Global Cities Program is not that old. It was established in 2018, as a response to the observation that cities were no longer just local actors. Instead, they've become laboratories for global solutions, and also important actors in global governance. So at Cities, we analyze the emergence of cities in international relations from three perspectives. First, we tackle cities in global governance, institutions, processes, and also norms. Second, we tackle the environmental challenges that cities might face and here we deal, specifically and mainly with climate change. And thirdly, we analyzed cities and socioeconomic and digital challenges, which is the research I'm leading. In the sense specifically, I'm interested in the discussion of how technology impacts lives, and rights of people living in cities. So we're actively promoting digital rights and the idea of technological humanism, both of which lie at the core, we are doing at CIDOB.

1:40 If you are not familiar with this term, technological humanism has to do with pursuing digitalization strategies, in which the technology deployment respects human rights at the same time that it fosters social innovation. So it means going from a tech driven approach to digitalization to human driven. One of the projects I'm leading is the Global Observatory of Urban Artificial Intelligence. I will explain in a minute with more about it. And I guess this is also the reason that I'm here with you today. Okay, so with no further ado, let's move into the structure of the course, what we have today. The guiding thread of this presentation will be the idea of trustworthy service. So ethical artificial intelligence applied to human context. And how are we doing?

2:38 Well, I seek to respond to three main questions. What is urban artificial intelligence? Specifically, what is ethical urban artificial intelligence? Second, why does this matter for cities? So why cities should be thinking and promoting algorithmic tools to deploy their services and their solutions? Also what challenges and opportunities

AI ETHICS

does artificial intelligence bring to cities? And finally, how are cities responding to the challenge of moving towards a more ethical use of artificial intelligence? What principles, what regulations, what best practices are of importance here? So the what, the why and the how of ethical urban artificial intelligence. But I'd like to start by playing a segment of a short documentary film we produced at CIDOB, which we released a couple of weeks ago, and that you may find very soon in our YouTube channel as well, just to start reflecting on the big questions around the ethical use of artificial intelligence in cities.

4:09 [Documentary clip] Life is speeding up. We live under constant changes that can make us feel uncomfortable. It isn't very easy to assume the omnipresence of technology without understanding its scope and where it will lead us. And in the meantime, data about our lives are being collected and processed with the help of artificial intelligence. And what exactly is artificial intelligence? Do we know what the actual presence of AI in our cities means and how it affects us? What does it really mean to make decisions based on our data? What do AI based decisions imply? What risks does it pose and to what extent can it improve the lives of people living in cities? Trustworthy Cities: Ethics in the AI Era.

5:37 I hope we were able to respond to some of these questions in the presentation today. But for the moment, let's go into a definition. What is urban artificial intelligence? So we can define it as the collection, interpretation and analysis of urban data in order to support policy-related decision-making, as well as the development of solutions that are, or could be, used in an urban context. And I'd like you to see now, play the continuation of this video, some of the solutions are already being implemented in many cities worldwide.

6:20 [Documentary clip] There are currently several sorts of AI used to improve the management of life in cities around the world. They are used in several different fields. Let's see some examples. In traffic control and to improve mobility in cities such as London, Austin, or New Jersey. In chatbots, in order to enhance communication with citizens regarding different topics. For health inquiries in Helsinki. To find and compare parking prices in Buenos Aires. Or to manage digital procedures in Singapore. Artificial Intelligence is also used to improve waste management in the streets of different Dutch cities or in Melbourne, Australia. Another application is facial recognition used in cities such as London and in Madrid, where it has been pilot tested to pay in some public means of transport by using facial recognition. We also find AI systems to manage air quality forecast data, or to certify if our environment is subject to different kinds of pollution in cities such as Vilnius or Los Angeles. In cities like Barcelona, it is also used to manage petitions by citizens, and also to support the management of public space by monitoring the city's beaches' capacity.

7:39 So we've seen just some examples of how artificial intelligence can be applied to cities. But let's look now on some of the big questions around the application of this technology in our cities. For instance, where are the limits of the decisions we can allow artificial intelligence to make? Should we set any limit? How can we ensure that the algorithmic tools do not discriminate on the basis of gender, sexual orientation, and others? How can we ensure that fundamental democratic values such as freedom and equality, or democracy are not being jeopardized? What is the role of the private sector in the deployment and control of these new technologies? How can public administrations lead innovation regarding AI in the private sector, in order to guarantee ethical standards? And that's an important one, because we know that most of the solutions are being deployed in the private sector, right? So, there is an imbalanced relationship with administrations. And then who is responsible when AI fails? So the question of accountability, and this relates to some of the principles that we will discuss today.

AI ETHICS

9:09 So in the context of this debate, and these questions, a key issue, it's what cities, what tools do cities have to promote the ethical use and deployment of urban artificial intelligence solutions. And here we can speak mainly of three tools. First, has to do with local innovation ecosystems, so administrations can align the innovation ecosystems, with the values of the technological humanism, make sure that many many actors within their cities, private companies but also civil society organizations, academia, research, work to promote or as a core business, an ethical use of artificial intelligence. The more solutions we have in the sense, the more close we are to having a variety of options to deploy ethical artificial intelligence in cities. But then a second really, really important tool is procurement, public procurement. And you can work towards this direction by introducing, for instance, digital rights clauses in the tendering process. This is something that many cities around the world are doing. And there are even some open templates that are being circulated for cities interested in promoting similar policies. And the last one, the big one, regulation. And that's not always easy. Here we have from setting standards and principles to attracting strategies – AI strategies similar to the ones that have many national governments, but also international institutions.

11:02 And this is the next point I want to make, regulation. How we advance towards regulation of artificial intelligence at the local level. The problem is that, in fact, all cities are affected by the regulation for the tiers of government. In fact, very few have in place local regulations around unethical use of artificial intelligence. In fact, if we briefly look into how is AI being regulated globally, we see a fragmented scenario where to take in this regulation efforts, we need to take into account the efforts by national governments. So since 2016, there has been a certain proliferation of national declarations, frameworks and strategies seeking to regulate AI. But as I was mentioning, these efforts are still very fragmented. We lack a common framework. If we look specifically at the global level, some of the initiatives worth taking into account are the Recommendations on the Ethics of Artificial Intelligence by UNESCO, the OECD AI Principles, and also the upcoming AI Act of the European Union, which for the moment is the most advanced regulation, which provides a risk based framework to regulate artificial intelligence.

12:43 But the key question here is: where is the local level, right? And I like to mention two initiatives here that are connected. All these initiatives, in fact, are connected to the idea that no city can go alone when advancing in the definition of ethical AI principles suitable to the local level. Here, I'll refer to the importance of alliances of cities and discuss mainly two examples. One is Eurocities. Before that, I like to make a brief comment on city networks. City networks are really important actors. They are networking governance structures that facilitate cooperation among cities with other actors. They serve also as platforms to lobby for specific advocacy goals, but also important actors to support capacity building, knowledge exchange and also promote policy implementation at the local level. We will take the case of Eurocities. Eurocities is the major city network gathering the largest cities in Europe, it has more than 200 members. And in the context of artificial intelligence, it is interesting the work that they're doing because they're sort of like collecting the bills that cities may have on this on the upcoming Artificial Intelligence Act at the European level, and bringing the demands of the cities to the current negotiation, so as to make sure that this global, this sort of regional standard takes into consideration or is suitable also for the local level. And here you can see some of the questions that Eurocities is bringing to the discussion such as the impact the AI may have on local job markets, or the issue around the collection of data, because now that AI nets huge amounts of data, and most of this data will be collected and processed and used in cities.

AI ETHICS

15:01 But I'd like to discuss another initiative, which is the Cities Coalition for Digital Rights. The Cities Coalition for Digital Rights was created in 2018 by three leading cities: Barcelona, Amsterdam and New York. And it emerged out of the observation that cities need to enforce digital rights but that there was a vacuum. There was a gap in the understanding of what cities have of those digital rights. What does a digital rights mean in practice? But also there was a lack of commitment of many cities around the world around digital. So they seek to create momentum to force the commitment around the digital rights movement, but also to provide a space for capacity building around different cities to discuss in a sort of like safe environment, the challenges they face in their day to day running of the cities when implementing those digital rights framework. Today, the coalition is composed of 55 cities and here you have some examples. They're from all over the world from Bordeaux to Los Angeles, also Stockholm or Sydney. And the coalition is supported by some patterns: UN Habitat, the UN agency for human settlements, Eurocities that I just mentioned before, UCLG, and Metropolis which are again two big city networks.

16:39 Out of this collaboration emerges a Declaration of the Cities Coalition for Digital Rights, in which cities commit to promote the universal and equal access to the internet and digital literacy. Also to ensure or to work towards ensuring privacy and data protection and security. Also, they identified as a principle that has to do with digital rights, transparency, accountability, and non discrimination of data, content and algorithms. Third, participation, democracy, diversity and inclusion. And finally, open an ethical digital service standards. Those five principles or those five dimensions of the Declaration on Digital Rights are, as we will discuss in a minute, very relevant also for their commitments for the initiative regarding the regulation of artificial intelligence. So ethical artificial intelligence is closely connected in this sense to digital rights frameworks.

17:54 One of the initiatives of the Cities Coalition for Digital Rights was the Global Observatory of Urban Artificial Intelligence. And here, you have the website so you can access it. The Global Observatory for Artificial Intelligence is a joint project of CIDOB, so a think tank I'm belonging to, and the cities of Barcelona, Amsterdam and London, so three core members of the Cities Coalition for Digital Rights with the support of UN HABITAT.

18:37 The Global Observatory of Urban Artificial Intelligence has three main aims. First, research - so filling the research gaps in the development of standards on ethical urban artificial intelligence. Here we developed six principles that we'll discuss in a minute. The second is monitoring AI initiatives right? So knowing a little bit, what is the state of the art, what cities around the world are doing, what challenges they're facing? So it's sort of like tracking and, following the developments of this emerging field of urban artificial intelligence. And finally, dissemination. So it's not only contributing to fill research gaps, monitoring what is being developed, but also disseminating these practices, disseminating principles, practices, strategies, so as to inspire more cities around the world to promote similar initiatives.

19:46 So let's now look into ethical artificial intelligence principles I just mentioned. And here we find: fairness and non-discrimination; transparency and openness; accountability; safety and cybersecurity; privacy protection; and sustainability. You see these six principles are nothing new. They're commonly found in national government strategies and declarations, and also in some of the global frameworks I just mentioned. What we did, in collaboration with many cities of the Cities Coalition for Digital Rights through a set of workshops and meetings, was identify those principles that could be applied and relevant also to the local level. The next step is we have six principles, how can we implement them? What do those principles mean, in practice? And to do that, we thought

AI ETHICS

that it would be relevant to identify examples, best practices of ethical uses of artificial intelligence in cities around the world, so as to inspire us. To give us ideas of how can these principles could be implemented, translated into specific projects.

21:03 And here is where, it comes the idea of launching an Atlas of Urban Artificial Intelligence. This is a tool that we produced to serve the needs of both cities, but also researchers. And it sorts, it presents different initiatives on ethical use of artificial intelligence. You can here, sort for instance, by the size of the city, so you can compare cities with similar dimensions. Also, by type of initiatives, for instance, it is quite common for a city to start by a set of small projects that they come together into a policy and might eventually become a strategy. But also we have differing levels of implementation of initiatives from early stage to fully implemented one. And we have initiatives from a variety of sectors. We mentioned that at the beginning of this course urban artificial intelligence is being applied to all sorts of fields from mobility to social services. Here you can also track by the six principles I just mentioned. So if you're interested in seeing initiatives that have to do with accountability, you can do that. Also, for the moment, we have around 113 initiatives from 40 key cities that can be also seen in a list. So you have the map view, but also the list.

22:40 Let's look at a couple of examples. We have Barcelona, the city where I live. Here we have some interesting projects, such as the use of AI to monitor beach capacity, but also AI in social services or the use in communication channels. Barcelona actually is one of the few cities having an AI strategy. And here for each of the projects on this initiative, you have a description and then some basic information regarding the principles that have been used, the implementation phase, the timeline, what was this initiative being launched and some additional information. Another very active city is Amsterdam. Amsterdam actually has 17 projects with the city with more projects in our Atlas. And here you find some more examples on how can artificial intelligence be used in an ethical way. And you have, for instance, one of the key initiatives is the Algorithm Register, but you have also how to eradicate poverty or waste management. I invite you to have a look at that.

23:54 Let's go back again to the ethical principles and see how they can be translated into practice. First, what they mean and what key questions cities need to ask themselves when implementing those principles. So let's start by fairness and non-discrimination, which can be described as the ability of an algorithmic tool to not discriminate against an individual or group because of personal characteristics of membership a popular level of group such as ethnic or sexual minorities. And here we produce such as a checklist, I would say, that series can go through and see whether the tools, the project they have meet these principles. For instance, they need to, one key question in this regard is does the system have a clear task to perform or how has the task been chosen? Also citizen participation - have citizens participated in it in any way? What technology was used? Does it employ publicly available tools? Also, is there a strategy or a set of procedures to avoid bias in the AI system during the design? Does it include diversity and representativeness? Also, is there a mechanism for citizens to report discriminatory or biased practices? Those are good questions around fairness and non-discrimination.

25:30 Let's look now into transparency and openness, which can be defined as the ability of an external actor to investigate the relationship between two variables, and see through the model's architecture. It means examining the relationship between cause and effect, which is interpretability, and/or the significance of weighting, so explainability. In fact, this principle is connected to the question of how to understand what and why AI does whatever it does. So in the sense interpretable and explainable models of AI are as important as determining

AI ETHICS

responsibility. Some good questions here: Is there a mechanism for citizens to report potential abuses? How is the quality of the input assessed? Is the system traceable? So questions of traceability, transparency, explainability. Are the decisions of the system transparent? Is the system explainable? Do citizens understand the system's decisions? How do they communicate whether they understand them or not? What mechanism does the city use to provide information to residents about AI systems deployed in the public space? And finally, how does the city provide information about risks and the necessity of use?

27:03 Accountability. Accountability can be defined as the state of being responsible or answerable for a system, its behavior and its potential impacts. It refers to allocating responsibility for and demonstrating compliance with legal requirements and moral (ethical) principles. So essentially, the principle of accountability seeks to respond to who should be blamed when AI causes harm. For instance, who is responsible when a self-driving car crashes, or an intelligent medical service or device fails? So the key questions here is: is the system auditable? How frequently is a system audited? Who audits the system? Will third parties do so, or is it a task force from the local administration?

27:55 Let's now some examples from this, incorporating these three principles. Here, the most relevant example would be AI registers being developed by the Cities of Helsinki, of Amsterdam. Interestingly, they've done that collaboratively. So again, I refer to this idea of alliances and city networks as important spaces to advance on ethical use of artificial intelligence. So basically, an AI register is a window. It's for citizens to see sort of like how or to learn about how artificial intelligence is being used as part of cities services, right. And here you have the cases of health center chatbots, or rental chatbots that had to do with apartments in Helsinki. Similar algorithmic system is being deployed in Amsterdam, and here for each of, of this, like services that the city services that use algorithmic systems, the city commits to provide the following information: the data sets that have been used to train the system; the data processing procedures; also if the system includes non discrimination provisions; if there is some human oversight on the decisions; and also finally, risk management considerations.

29:34 Let's move to the next principle: privacy protection, which in fact, is the most basic digital right, right. It's one of the most prominent digital rights for sure, and an essential component of unethical use of algorithmic tools. Actually, the administration should pay attention to the level of inclusiveness and assess the potential impact, direct or indirect, of the tool on the citizen's right to privacy. So, how can we refine the privacy protection is ensured when cities guarantee anonymization of data, create security against deanonymization, guarantee that data is always available and adhere to the right of explanation. Some key questions include: what data has been used to train the algorithm? Where is that data from? What is the relation between or behind the chosen data? Which measures have been put in place to achieve privacy by design systems? Who takes care of data, so, the data management? Whose data? How is this data processed? What are some safeguards? Does the system comply with data protection laws? How? What is the impact of the system on the individual's right to privacy, intimacy, dignity and integrity?

31:05 And here an example I want to share with you is the city of Montreal who deployed a snow level detection system that is basically AI. What AI does here is it helps the city to prevent fraud made by snow removal companies by verifying the level of snow in each reported dump truck. So the model basically provides the level of snow in percentage using a picture as input. The cameras can only identify snow, so no plate numbers are identified in the process. And this is the privacy element we just discussed.

AI ETHICS

31:45 Safety and cybersecurity. So, algorithmic decisions must be resilient to cyberattacks and adversarial threats that have the objective of altering their functions. In this sense, safety and cybersecurity refer to the process of ensuring, sorry, the process of ensuring a high degree of cybersecurity and technical robustness in the algorithmic decision systems to make them resilient to cyberattacks and adversarial threats that have the objective of altering their functions. Again, key questions here: What are the potential risks to AI systems? Are there measures in place to mitigate potential negative consequences? Is there a mechanism to monitor and evaluate the performance of the AI system? Is the AI system reliable and stable? How does performance affect its behavior? How is the system updated? Is the AI system reliable and stable? As I mentioned, has the administration considered how to keep the system up to date, so maintenance? Does the system design consider potential vulnerabilities related to data poisoning, model evasion, or inversion?

33:06 Sustainability. Sustainability has to do with efforts of finding new ways to develop AI systems in the most environmentally friendly way possible. So this principle is linked to the fact that the use of AI systems can negatively impact global efforts to fight climate change, as AI relies on storage and data centers that are highly energy consuming. And so it's also linked to an increase in electronic waste generation. Key questions: What are the potential environmental impacts of the design and deployment of the AI system? Does the system consider the whole lifecycle of the system or only its design? How can cities reduce, compensate for, and mitigate these impacts? What is the role of the city during this data lifecycle? Do cities have urban data centers? What is the impact of the system on the local communities, specifically on local economies, social interaction, resident-administration interaction, people's rights and freedoms etc?

34:26 And here, I want to mention that the cities that have been more advanced in regulating AI are those having an AI strategy in place, for instance, the “Government measure for analytical algorithms and their strategy for an ethical promotion of artificial intelligence” of the city of Barcelona. It's an example but also the strategy of Amsterdam. But also London's emerging Tech Charter, or the artificial intelligence strategy of Buenos Aires. All of these strategies include notions of cybersecurity and safety, and also sustainability that I just explained. So if you're interested in learning about more projects that include an ethical dimension or an ethical consideration on the deployment of artificial intelligence solutions in an urban context, I invite you to check out our Atlas, where you can have much more examples. Also, you have here the link you can access. It's free and available on our website. And above all, I ask you also to contribute if you know of any experience, any project, initiative, strategy that is not included in the Atlas to contribute to help us with this global effort. Well, thank you for your attention. Here, you have my contact details, very happy to respond to any questions you may have. Thank You.